

How to create a FLEX system (Part 2) ...more specifically

 flexonsbd.blogspot.com/2018/06/flex.html

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How to create a FLEX system... in more detail

This explains the previous "How to Create a FLEX System" (published 2018/05/04) in more detail.

Most of the nearly 100 files in the folder FLEX6809 are older versions of FLEX such as 9.0 and 2.9, and their structure is slightly different from 3.01, so the procedure explained last time cannot be applied as is. (The structure of 9.0 and 2.9 is similar to FLEX2 for 6800.)

The required version V3.01 is only found in FLEXSSB.DSK.

FLEX.COR is not in this DSK, but FLEX.SYS is in the form of DRIVER.BIN and CONSOLE.BIN appended to the end of FLEX.COR, so if you remove these, you will get FLEX.COR. If you append the DRIVER.BIN and CONSOLE.BIN for this system to the obtained FLEX.COR, you will get FLEX.SYS.

Specific steps

(All numbers below are in hexadecimal notation.)

(1) Obtain FLEX.COR

```
003F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00400 00 06 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00410 45 52 52 4F 52 53 00 00 53 59 53 00 00 01 01 01 ERRORS..SYS....
00420 09 00 09 02 00 0C 0E 57 46 4C 45 58 00 00 00 00 .....WFLEX...
00430 53 59 53 00 00 01 0A 01 21 00 18 00 00 0C 0E 57 SYS.....!.....W
00440 50 52 49 4E 54 00 00 00 53 59 53 00 00 01 22 01 PRINT...SYS...".
00450 22 00 01 00 00 0C 0E 57 43 41 54 00 00 00 00 00 ".....WCAT....
00460 43 4D 44 00 00 01 23 02 01 00 03 00 00 0C 0E 57 CMD...#.....W
00470 43 4F 50 59 00 00 00 00 43 4D 44 00 00 02 02 02 COPY...CMD....
00480 06 00 05 00 00 0C 0E 57 44 45 4C 45 54 45 00 00 .....WDELETE..
00490 43 4D 44 00 00 02 07 02 08 00 02 00 00 0C 0E 57 CMD.....W
004A0 4C 49 53 54 00 00 00 00 43 4D 44 00 00 02 09 02 LIST...CMD....
004B0 08 00 03 00 00 0C 0E 57 50 00 00 00 00 00 00 00 .....WP.....
004C0 43 4D 44 00 00 02 0C 02 0C 00 01 00 00 0C 0E 57 CMD.....W
004D0 41 53 4E 00 00 00 00 00 43 4D 44 00 00 02 0D 02 ASN...CMD....
004E0 0D 00 01 00 00 0C 0E 57 52 45 4E 41 4D 45 00 00 .....WRENAME..
004F0 43 4D 44 00 00 02 0E 02 0E 00 01 00 00 0C 0E 57 CMD.....W
00500 00 07 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00510 41 50 50 45 4E 44 00 00 43 4D 44 00 00 02 0F 02 APPEND..CMD....
00520 11 00 03 00 00 0C 0E 57 42 55 49 4C 44 00 00 00 .....WBUILD...
00530 43 4D 44 00 00 02 12 02 12 00 01 00 00 0C 0E 57 CMD.....W
00540 45 58 45 43 00 00 00 00 43 4D 44 00 00 02 13 02 EXEC...CMD....
00550 13 00 01 00 00 0C 0E 57 53 41 56 45 00 00 00 00 .....WSAVE....
```

Image 1 Directory

If you open FLEXSSB.DSK with a binary editor and look at the directory starting at address 410, you will see that FLEX.SYS is a sector track with 18 sectors, from sector number 01,0A to 01,21.

The addresses are 2D00 to 44FF, so copy and extract them and save them as FLEX.SYS.

```
02C00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
02D00 01 08 00 01 02 CC 00 3B 08 18 3A 00 00 04 00 00 .....:..:
02D10 00 FF 1B 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
02D20 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
02D30 00 00 00 BF FF 00 00 01 00 00 00 00 00 00 00 00 .....
02D40 00 00 00 02 CC 49 01 80 02 CC 4E 5A 2B 2B 2B 04 .....1...NZ+++
02D50 3F 3F 3F 04 57 48 41 54 3F 04 43 41 4E 27 54 20 ???..WHAT?.CAN'T
02D60 54 52 41 4E 53 46 45 52 04 4E 4F 54 20 46 4F 55 TRANSFER..NOT FOU
02D70 4E 44 04 44 49 53 4B 20 45 52 52 4F 52 20 23 04 NO..DISK ERROR #.
02D80 44 52 49 56 45 53 20 4E 4F 54 20 52 45 41 44 59 DRIVES NOT READY
02D90 04 47 45 54 00 D2 0F 4D 4F 4E 00 D3 4B 00 27 10 .GET...MON..K..
02DA0 03 E8 00 64 00 0A 02 CC 00 01 39 02 CC D8 01 39 ...d.....9...9
02DB0 02 CC E4 01 39 02 CC F8 C4 01 00 00 00 00 00 00 ...9.....
02DC0 00 7E CD 57 7E CD 67 7E CD B1 7E CD 09 7E CD CC ..N-E-..A-+
02DD0 7E CD 0F 7E CD 12 7E CE FB 7E CF 41 7E CE 2B 7E ..-..-..A-+
02DE0 CE 81 7E CF ED 7E CE B5 7E D0 0E 7E CD EA 7E D0 ..-..-..-..-
02DF0 38 7E D1 A2 7E D0 EB 7E D3 49 7E CF 86 7E CF D4 6-..-..1-..-
02E00 01 0C 00 02 7E D2 7F 7E D1 2F 7E CF D0 7E D1 75 ...-..-..-..u
02E10 7E D3 29 7E CD 4E 7E CE 05 7E CE 05 10 CE C0 7F ..)'.N'. ....
02E20 7F CC 11 BD D4 00 7F CC 28 BD D3 FD 10 CE C0 7F .....(.....
02E30 BD DE 18 8E CD 03 BF CC 16 8E C7 00 AF 9F D3 E9 .....
02E40 BE D3 E7 AF 9F D3 EB 8E CC F8 BF CC 31 7F CC 34 .....1..4
02E50 7F CC 4C 8D 5A B6 CC 11 B1 CC 02 26 05 7C CC 15 ..L.Z.....&.]..
02E60 20 14 7D CC 28 10 26 05 97 BD D4 03 26 B2 8E CC ..]..&.....&..
02E70 4E BD CE 81 8D 7A BD D0 D4 81 00 27 E5 8E C8 40 N....z.....'...@
```

Image 2: Beginning of FLEX.SYS

Looking at the end of FLEX.SYS, the last two sectors appear to be independent.

The sector at address 4300 is DRIVER.BIN, and the sector at 4400 is CONSOLE.BIN.

(DRIVER.BIN is simple because its actual content is in other parts.)

FLEX.COR is obtained by extracting and saving up to 42FF, excluding these two sectors.

```

341E0 39 BE D4 0B A6 03 4C 81 04 24 0F A7 03 26 05 BD 9.....9..U..d..
341F0 39 BE D4 0B A6 03 4C 81 04 24 0F A7 03 26 05 BD 9.....L..$.3..
34200 01 20 00 16 DE 0F 20 03 8D DE 12 25 E8 39 C8 10 .....X.9..
34210 1A 01 39 00 00 00 00 00 00 00 00 00 00 00 00 ..8.....
34220 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34230 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34240 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34250 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34260 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34270 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34280 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34290 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
342A0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
342B0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
342C0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
342D0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
342E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
342F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34300 01 21 00 17 02 DE 00 1E 7E E8 10 7E E8 13 7E E8 .....
34310 16 7E E8 19 7E E8 1C 7E E8 1F 7E E8 22 7E E8 25 .....
34320 7E E8 28 7E E8 2B 00 00 00 00 00 00 00 00 00 .....
34330 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34340 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34350 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34360 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34370 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34380 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34390 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
343A0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
343B0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
343C0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
343D0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
343E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
343F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
34400 00 00 00 18 02 D3 E5 18 D3 88 D3 D6 F3 C2 F3 C8 .....
34410 D3 CF D3 C8 D3 B5 FB 18 D3 70 D3 AA D3 98 D3 97 .....
34420 02 D3 70 6D BE F3 E2 BF D3 E3 30 01 BF D3 E1 86 .....
34430 13 A7 9F D3 E3 86 16 A7 9F D3 E3 39 A6 9F D3 E3 .....
34440 84 01 27 F8 A6 9F D3 E1 84 7F 39 8D EF 34 02 A6 .....
34450 9F D3 E3 84 02 27 F8 35 02 A7 9F D3 E1 39 34 02 .....
34460 A6 9F D3 E3 84 01 35 02 39 BE D3 DF 86 FF A7 84 .....
34470 86 3C A7 01 86 8F A7 84 A6 84 86 8D A7 01 39 86 .....
34480 04 20 02 86 8F A7 9F D3 DF 39 A6 9F D3 DF 7E C7 .....
34490 00 02 D3 DF 06 F7 F2 F7 E8 F7 E8 16 CD 00 00 00 .....
344A0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

```

Image 3: End of FLEX.SYS

(2) Prepare Driver.bin and Console.bin

First, go to GitHub as introduced on the previous page, click the Clone or download button, and select Download ZIP to download 6809_6800_FLEX-master.zip. Extract this file. Assemble DRIVER.TXT and CONSOLE.TXT from this file using the included 6809AsmWin.exe to obtain DRIVER.BIN and CONSOLE.BIN.

(3) Convert the bin file to packet format

The obtained BIN file cannot be directly placed into the DSK file.

As mentioned in "Details of Single Board Computers and the FLEX System" (published May 4, 2018), a plain BIN file cannot exist in the FLEX system, so it must be converted to packet format. To do this, prepare an empty DSK file (BLANK.DSK), set it in the drive of FlexDrvWin.exe, and simply drag and drop the BIN file onto it. This will automatically convert it to packet format, and sector information will be added to the beginning of each sector.

(3-1) Convert DRIVER.BIN

```

31FF0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32000 01 02 00 01 02 DE 00 C4 7E DE 65 7E DE DB 7E DE .....e...
32010 22 7E DE 24 7E DE 27 7E DE 39 7E DE 39 7E DE 49 .....$...9..i
32020 7E DE 48 7E DE 61 00 00 00 00 5F 39 8D 01 39 A6 .....F.a...9..
32030 03 81 03 23 05 C6 0F 1A 01 39 B7 DE 1E 5F 1C FE .....#...9...
32040 39 A6 03 81 03 23 05 C6 80 1A 01 39 5F 1C FE 39 9....#...9...9
32050 39 7F DE 1E 86 43 B7 F0 CC 12 86 55 B7 F0 CC B6 9....C...U...
32060 F0 CD B6 F0 CC 47 25 F7 39 5F 1C FE 39 7F DE 1F .....GX.9...9...
32070 34 02 86 52 17 00 D0 8D 63 B6 DE 1E 17 00 D5 8D 4..R...c...
32080 5B 35 02 17 00 CE 8D 54 1F 98 17 00 C7 8D 4D B6 [5....T.....M..
32090 DE 1F 17 00 BF 17 00 A4 81 15 27 D1 81 1B 26 04 .....&.....
320A0 C6 10 5D 39 81 06 26 ED 17 00 A9 BF DE 20 BE DE ..]9..&.....
320B0 20 5F 7F DE 1F 17 00 84 A7 80 8D 20 5A 26 F6 17 .....Z&...
320C0 00 7A B1 DE 1F 27 07 86 15 17 00 88 02 DE C4 A5 ..z.....
320D0 20 E0 86 06 17 00 81 17 00 68 81 06 26 F9 5F 39 .....f...&..9
320E0 B6 DE 1F B7 DE 1F 39 7F DE 1F 34 02 86 57 8D 68 .....9...4..W.h
320F0 8D EE B6 DE 1E 8D 61 8D E7 35 02 8D 5B 8D E1 1F .....a..5..[...
32100 00 00 00 02 98 8D 55 8D DB B6 DE 1F 8D 4E 8D 34 .....U.....N.4
32110 81 15 27 D7 81 1B 26 04 C6 10 5D 39 81 06 26 EE .....&...39..&..
32120 BF DE 20 BE DE 20 5F 7F DE 1F A6 8D 8D 2E 8D B4 .....&.....
32130 5A 26 F7 B6 DE 1F 8D 24 8D 0A 81 15 27 E5 81 06 Z&.....$....
32140 26 F6 5F 39 34 04 C6 55 86 15 B7 F0 CC 12 B6 F0 &_94..U.....
32150 CC 47 24 FA B6 F0 CD F7 F0 CC 35 84 34 04 C6 55 ..G$.5...4..U
32160 34 02 86 15 B7 F0 CC 12 B6 F0 CC 47 47 24 F9 35 4.....G$.5
32170 02 B7 F0 CD F7 F0 CC 35 84 00 00 00 00 00 00 .....5.....
32180 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32190 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
321A0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
321B0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
321C0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
321D0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
321E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
321F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32200 00 00 00 01 02 D3 E5 57 D3 7D D3 7D FF FF FF .....W.p.]...
32210 D3 7C D3 7C D3 7C F8 37 D3 99 D3 8F D3 8D D3 7E .|.|.|.7.....
32220 B6 F0 BC 84 01 27 F9 B6 F0 BD 84 7F 39 3B 8D F0 .....9;...
32230 34 02 B6 F0 BC 84 02 27 F9 35 02 B7 F0 BD 39 34 4.....5...94
32240 02 B6 F0 BC 84 01 35 02 39 86 43 B7 F0 BC 12 86 .....5.9.C....
32250 15 B7 F0 BC B6 F0 BD B6 F0 BC 84 01 26 F6 39 16 .....&.9..
32260 CD 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

```

Image 4: DRIVER.BIN and CONSOLE.BIN

Drag and drop DRIVER.BIN into the file contents display box of FlexDrvWin.exe.

It will ask for the starting address, so select "No" to exit, set the value in the Start Adr box of the binary file to 0xDE00, and drop it again. The execution address can remain at 0x0000, so answer "Yes" and DRIVER.BIN will be saved in packet format starting from the specified address.

Open the saved BLANK.DSK with a binary editor, and you will see that it is saved in 2 sectors starting from address 2000, so copy and extract this and save it as DRIVER_DSK.BIN.

(3-2) Convert CONSOLE.BIN

```

021F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
02200 00 00 00 01 02 D3 E5 57 D3 70 D3 7D FF FF FF FF .....W.p.]...
02210 D3 7C D3 7C D3 7C F8 37 D3 99 D3 8F D3 80 D3 7E .|.|.|.7.....~
02220 B6 F0 BC 84 01 27 F9 B6 F0 BD 84 7F 39 3B 8D F0 .....9;...
02230 34 02 B6 F0 BC 84 02 27 F9 35 02 B7 F0 BD 39 34 4.....'.5...94
02240 02 B6 F0 BC 84 01 35 02 39 86 43 B7 F0 BC 12 86 .....5.9.C...
02250 15 B7 F0 BC B6 F0 BD B6 F0 BC 84 01 26 F6 39 16 .....&.9...
02260 CD 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
02270 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
02280 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
02290 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
022A0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
022B0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
022C0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
022D0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
022E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
022F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
02300 00 00 00 01 02 D3 E5 18 D3 70 D3 7D FF FF FF FF .....p.]...
02310 D3 7C D3 7C D3 7C F8 37 D3 99 D3 8F D3 80 D3 7E .|.|.|.7.....~
02320 02 D3 70 3F B6 F0 BC 84 01 27 F9 B6 F0 BD 84 7F ..p?.....'.
02330 39 3B 8D F0 34 02 B6 F0 BC 84 02 27 F9 35 02 B7 9;..4.....'.5..
02340 F0 BD 39 34 02 B6 F0 BC 84 01 35 02 39 86 43 B7 ..94.....5.9.C.
02350 F0 BC 12 86 15 B7 F0 BC B6 F0 BD B6 F0 BC 84 01 .....
02360 26 F6 39 16 CD 00 00 00 00 00 00 00 00 00 00 &.9.....
02370 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
02380 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

```

Image 5: Editing CONSOLE.BIN

The method is the same as above, but there are some points to note. As you can see from CONSOLE.TXT, the first half is the sequence of I/O vector values starting from D3E5, and the second half is the actual I/O program starting from D370. If you just throw this into FlexDrvWin.exe as is, the whole thing will start from D3E5.

(Top of Image 5)

You could process them separately and then combine them at the end, but since the size isn't that big, we'll just modify it manually and throw in CONSOLE.BIN with the starting address set to 0xD3E5 and the execution address set to 0xCD00.

(The execution address is specified to add 16,CD,00 to the end.)

Then change the following two places. (Two red-framed areas at the bottom of Image 5)

- Since the vector part is 18 bytes, change 57 after 02,D3,E5 to 18.
- Insert 02,D3,70,3F after the 24 bytes starting from D3,70.

Copy and extract this one sector and save it as CONSOLE_DSK.BIN.

(4) Concatenate DRIVER_DSK.BIN and CONSOLE_DSK.BIN to FLEX.COR

Prepare a new BLANK.DSK and copy FLEX.COR to 2000-35FF.

Copy DRIVER_DSK.BIN to the two sectors after that, and CONSOLE_DSK.BIN to the sector after that.

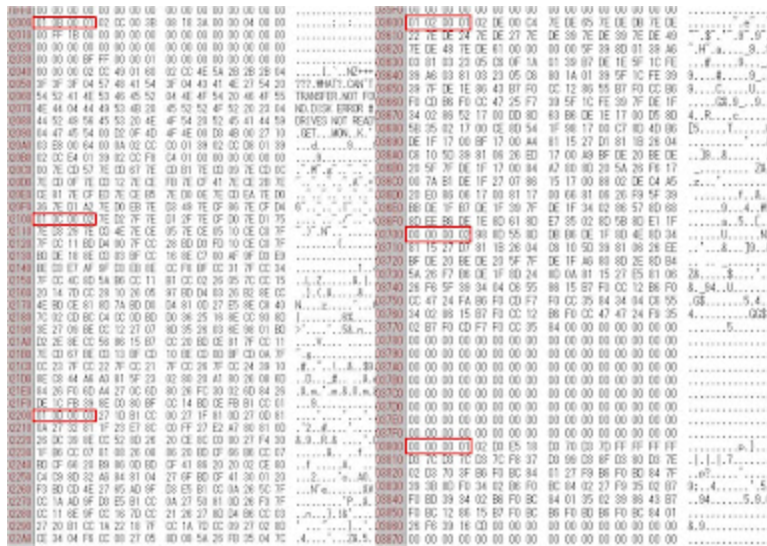


Image 6: The beginning (left) and end (right) of FLEX.SYS.

(5) Rewriting Sector Information

Now that all sectors are gathered, we will organize the first 4 bytes of sector information for each sector.

The first 4 bytes of each sector starting from address 2000 are:

I will rewrite it as follows.

(6) Rewrite the SIR (System Information Record) information for sectors 00,03

The unused area starting sector from offset 1D in sectors 00,03 (address 200) is rewritten to 01,1A, and the remaining sector count from offset 21, 09,E0, is rewritten to 09,C7.



Image 7: SIR information

(7) Rewrite directory information in sector 00,05

Write the following 24 bytes from offset \$10 in sector 00,05 (address \$400):

File name (8 bytes), extension (3 bytes), attribute (1 byte, usually 00), unused (1 byte, 00),

start Trk,Sct (2 bytes, 01,01), end Trk,Sct (2 bytes, 01,19), size (2 bytes, 00,19), 2 bytes (00,00), date (3 bytes, month, day, year in binary).

00400	00 06 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00410	46 4C 45 58 00 00 00 00	53 59 53 00 00 01 01 01	FLEX....SYS.....
00420	19 00 19 00 00 04 1A 12	00 00 00 00 00 00 00 00	
00430	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00440	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00450	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00460	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00470	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00480	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00490	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
004A0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	

Image 8 directory

(8) Finally

The base of the system disk is now complete. All that remains is to find the DSK file containing the necessary CMD files from SWTPCemu, set it to the drive of FlexDrvWin.exe, and copy it to the system disk.

For example, if the DSK containing COPY.CMD is set to drive 2 and the DSK containing the CMD files you want to copy is set to drive 1, then

2.COPY (sp) 1 (sp) 0 (sp) .CMD (sp is a space)

will copy all CMD files to drive 0.

For details, please refer to the documentation in the TSC folder within the Documentation folder.

(9) Other

- The first two sectors of the system disk, 00,01 and 00,02, must contain the BootLoader. The FLEX command in ASSIST09 has the function to read these two sectors, set them to C100, and transfer execution. (The Trk,Sct numbers of the first sector of FLEX.SYS are written at offsets 05 and 06 of this BootLoader. The FLEX LINK command searches for the SYS file and writes its location here. This ensures that it can boot correctly no matter where the SYS file is located on the DSK.)

Changing the value of CC07 from 00 to 08 will erase the deleted characters when the BS key is pressed, making it easier to use. To do this, change the offset 0F (address 200F) from the beginning of FLEX.COR to 08.

Upload the BLANK.DSK, DRIVER_DSK.BIN, and CONSOLE_DSK.BIN files containing the BootLoader necessary for the above operations to [Microsoft OneDrive](#) with the filename MakeFLEX.